Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 14)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows some features of Earth, Mars and Venus.

Table 1

Feature	Venus	Earth	Mars
Mean distance from Sun / million km	108	150	228
Mass relative to Earth	0.82	1.0	0.11
Atmospheric pressure relative to Earth	92	1.0	0.006
Mean surface temperature / °C	460	15	-60
Magnetic field	Very weak	Strong	Very weak
Rotation period / hours	5 800	24	25

0 1 . 1 Use Table 1 to explain why liquid water cannot exist on the surface of Mars.

[3 marks]

0 1 . 2 Explain how the Earth's mass and its distance from the Sun created suitable conditions for life.

[3 marks]

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0 1 . 3 Explain the importance of the Earth's magnetic field and its speed of rotation for living organisms.

[2 marks]

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0 1 . 4 Explain how the tilt of the Earth's axis affects living organisms.

[2 marks]

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10

0 2 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 2 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 2 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

1 .....

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0 3 . 1 Explain what is meant by **biomimetics** and describe **three** examples.

[4 marks]

0 3 . 2 Explain why many plants produce chemicals that are useful as human medicines, and give **two** examples.

[3 marks]

0 3 . 3 Explain why the discovery of medicines is a strong argument for conserving whole habitats rather than individual species.

[2 marks]

0 3 . 4 Give **one** example of a wildlife species used in physiological research and the human health issue it helps to understand.

[1 mark]

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0 4 The Hawaiian goose (nene) was reduced by hunting, habitat loss and introduced predators to about 30 birds. A captive breeding programme started with just seven individuals and has increased the wild population to over 2000.

0 4 . 1 Explain why starting the programme with seven individuals caused problems, and describe how these problems are managed.

[4 marks]

0 4 . 2 Explain **three** reasons, other than gene pool size, why species kept in captivity may fail to breed.

[3 marks]

0 4 . 3 Describe how artificial insemination and embryo transfer increase breeding success in captive mammals.

[3 marks]

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0 5 . 1 Explain how each of the following agricultural changes affects wildlife habitats.

[6 marks]

Drainage of wet grassland

Application of fertiliser to a species-rich meadow

Removal of hedgerows to enlarge fields

0 5 . 2 Explain why some farmland habitats are valuable for wildlife conservation.

[2 marks]

0 5 . 3 Describe how agriculture contributes to climate change.

[2 marks]

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0 6 . 1 Define each of the following ecological terms and give an example.

[8 marks]

Species

Ecological niche

Community

Biome

0 6 . 2 Explain the difference between an ecosystem and a biome.

[2 marks]

10

0 7 . 1 Complete Table 1 to compare natural systems with typical linear human systems.

[4 marks]

Table 1

Feature	Natural systems	Linear human systems	
Energy source			
Use of materials			
Nature of wastes			
Temperature of processes			

0 7 . 2 Explain why natural systems can be sustained using a relatively small number of elements.

[2 marks]

0 7 . 3 Explain why the human use of minerals makes them difficult to recover and re-use.

[2 marks]

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0 7 . 4 Explain why toxic metals in electronic products cause problems that the wastes of natural systems do not.

[2 marks]

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0 8 . 1 Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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0 8 . 2 Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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0 8 . 3 Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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5

0 9 . 1 Describe how DNA databases are used to control the illegal trade in tropical timber and ivory.

[3 marks]

0 9 . 2 Explain the difference between radio / GPS tracking and acoustic tracking and state when each is used.

[2 marks]

0 9 . 3 Give **three** types of carrier system used to move sensors to study areas and one advantage of drones for ecological monitoring.

[3 marks]

0 9 . 4 Explain how motion-sensitive cameras and CCTV reduce disturbance to wildlife compared with direct observation.

[2 marks]

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1 0 . 1 Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

1 0 . 2 Explain why growing coral reefs help to resist climate change.

[2 marks]

1 0 . 3 Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.

[25 marks]

OR

1 1 . 2 Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS